The University of Faisalabad

Network Model

Protocols and Standards, Internet Standards

LAYERED TASKS: Sender, Receiver, and Carrier.

THE OSI MODEL: Layered Architecture ,

Peer-to-Peer Processes,

Encapsulation

Lecture # 7 and 8

Week# 4

Data Communications and Networking, Fourth Edition, Behrouz A. Forouzan

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Protocol

- **Protocols and Standards**

What is a Protocol?

- A **protocol** is a **set of rules** that computers follow to communicate with each other.

Protocol = Language of computers

Example

- Just like humans use **English or Urdu** to talk, computers use protocols like:
- HTTP
- TCP
- IP

Without protocols, computers **cannot understand each other**.

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Standards

- **What are Standards?**
- **Standards** are **official rules** that ensure:
- Devices from different companies work together
- Communication is reliable and correct

Example

- Laptop (HP) communicating with server (Dell)
- Mobile phone connecting to Wi-Fi router

This works because everyone follows the **same standards**.

Easy Analogy

- Traffic rules:
- Red = Stop
- Green = Go
  Everyone follows the same rules → no accidents

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Internet Standards

What are Internet Standards?

- Internet standards are **rules specifically made for the Internet**.

They define:

- How data is sent
- How websites work
- How emails are delivered

Examples of Internet Standards

- **HTTP** → Web browsing
- **FTP** → File transfer
- **SMTP** → Email sending
- **IP** → Addressing

Simple Example

- When you open **google.com**, it works because:
- Browser
- Server
  Both follow **Internet standards**

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Layered Tasks: Sender, Receiver, and Carrier

Communication is divided into **layers**, and each layer has a **specific job**.

<u>Sender</u>

- Prepares data
- Breaks data into packets
- Adds address information

Example:
You send a WhatsApp message.

<u>Carrier</u>

- Carries data through the network
- Uses cables, Wi-Fi, fiber, etc.

Example:
Internet cables and Wi-Fi signals

<u>Receiver</u>

- Receives data
- Reassembles packets
- Shows message to user

Example:
Your friend receives and reads the message.

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Layered Architecture

- **Layered Architecture (OSI Model Concept)**

What is Layered Architecture?

- Layered architecture means:
- Each layer performs **one specific function**
- Layers work **independently**
- **Example**
- If **internet is slow**:
- Physical layer → cable issue
- Network layer → IP problem

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The OSI Model

The **OSI Model** is a **7-layer reference model** that explains **how data moves from sender to receiver**.

OSI = *Open Systems Interconnection*

- **Why OSI Model?**
- Makes networking easy to understand
- Divides complex communication into steps

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7 Layers of OSI

Layer	Purpose
7. Application	User interaction (Browser, Email)
6. Presentation	Format & encryption
5. Session	Session management
4. Transport	Reliable delivery (TCP/UDP)
3. Network	Routing & IP addressing
2. Data Link	MAC address & framing
1. Physical	Cables, signals

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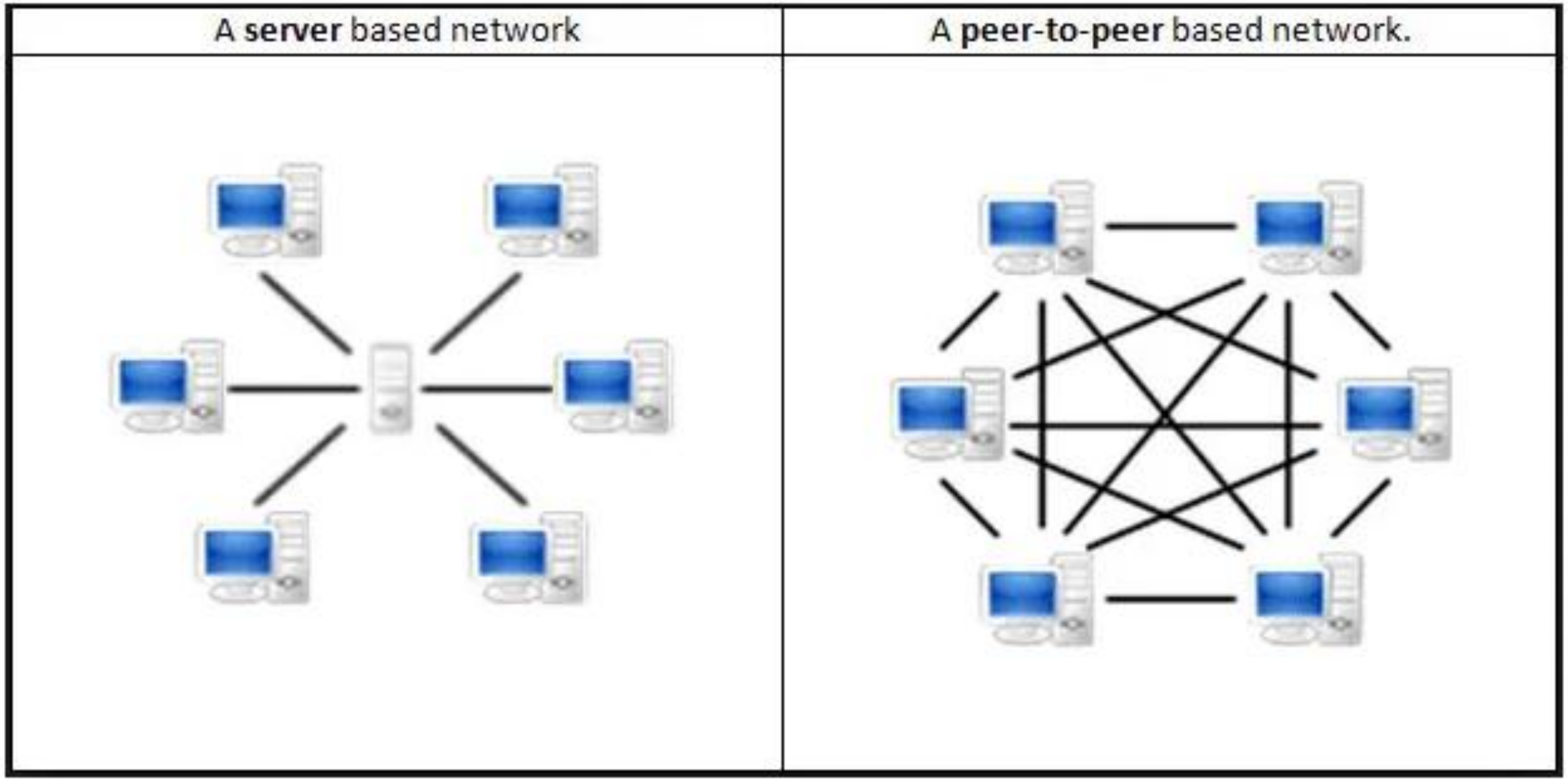
Peer-to-Peer Processes

- **What is Peer-to-Peer (P2P)?**
- In **Peer-to-Peer, both devices act as sender and receiver**.

No central server needed



Examples

- File sharing apps
- Bluetooth file transfer
- Torrent download

Easy Example

- Two friends:
- Both can talk
- Both can listen
  No boss in between

Encapsulation

What is Encapsulation?

- **Encapsulation** means **wrapping data with extra information** as it moves from the **sender to the receiver**.
- **Encapsulation = Packing data layer by layer before sending**
- When you send a **message or data**:
- **Step 1: Application Layer**
- Creates the **actual message**
- Example: “Hello”
- **Step 2: Transport Layer**
- Adds **port number**
- Helps deliver to correct app

Step 3: Network Layer
Adds **IP address**
Finds the path
Step 4: Data Link Layer
Adds **MAC address**
Helps deliver in local network
Step 5: Physical Layer
Converts data into **signals**
Sends through cable or Wi-Fi
Like courier carrying the box

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Encapsulation

- **WhatsApp Example**

You send **“Hi”** on WhatsApp:

- Message created
- Port added (WhatsApp app)
- IP added (friend’s phone)
- MAC added (local delivery)
- Data sent as signals

This full wrapping = **Encapsulation**

Encapsulation Through OSI Layers

```
Application Layer
        ↓
Presentation Layer
        ↓
Session Layer
        ↓
Transport Layer → Segment
        ↓
Network Layer → Packet
        ↓
Data Link Layer → Frame
        ↓
Physical Layer → Bits
```

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Encapsulation Flow

- **Segment (Transport Layer)**
- At the **Transport Layer**, data is divided into small parts called **segments**.
- This helps send data **reliably and in order**.
- **Example:**
  You send a large file by email. The system **breaks the file into smaller segments** so it can be transmitted easily.

- **Packet (Network Layer)**
- At the **Network Layer**, the segment becomes a **packet**.
- The packet contains the **IP address of sender and receiver** so the data can find the correct path across networks.
- **Example:**
  When you open a website, packets travel through **routers on the internet** using IP addresses

Encapsulation Flow

Frame (Data Link Layer)

- At the **Data Link Layer**, the packet becomes a **frame**.
- The frame contains **MAC addresses** of the sender and receiver devices.
- **Example:**
  Inside a **Wi-Fi or LAN network**, the router uses MAC addresses to send the frame to the correct computer.

- ## Bits (Physical Layer)

- At the **Physical Layer**, the frame is converted into **bits (0s and 1s)**.
- These bits travel through **network cables, fiber optics, or wireless signals**.
- **Example:**
  In an Ethernet cable, data moves as **electrical signals representing 0 and 1**.

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THANK YOU

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